Hong Lang

Postdoctoral Research Associate · Civil & Environmental Engineering, University of Illinois Urbana-Champaign

+1 217-250-7866 | honglang@illinois.edu | Hong Lang | Google Scholar | ORCID

HIGHLIGHT

- ***Funding:*** PI on seven funded projects totaling more than $1.0 M; contributed to 11 additional proposals, with $2.17M awarded.
- ***Research Output:*** Published 43 peer-reviewed journal articles and 20 conference papers; contributed to seven patents, two inspection standards, and open pavement datasets.
- ***Asphalt Material***: Developed and deployed the AI-enabled Optimal Mix Design Tool, used by more than 2,000 visitors from 20+ countries and the uncertainty-quantification-based Asphalt Binder Selection Guidance Tool.
- ***Mechanics:*** Developed an FEA-trained, physics-informed generative model for three-dimensional tire–pavement contact-stress prediction in 1.56 seconds.
- ***Sensing/NDE:*** Designed vehicle-mounted 3-D laser and LiDAR pavement-inspection systems and contributed to contact-less friction-estimation and real-time GPR systems.
- ***AI/ML:*** Developed physics-informed and transfer-learning models for asphalt mixture performance, tire–pavement stress, pavement distress, and friction; evaluated uncertainty and transferability across materials, sensors, and regions.
- ***Data Systems:*** Leading the modernization of the Illinois Pavement Feedback System, using four decades of records to support treatment selection and performance feedback.

ACADEMIC APPOINTMENTS

University of Illinois Urbana-Champaign — *Champaign, IL*
Postdoctoral Research Associate at Illinois Center for Transportation — *2024 – Present*

- Advisor: Professor Imad L. Al-Qadi

Tongji University — *Shanghai, China*
Research Associate Professor at the Transportation Research Institute — *2024 – 2024*

Tongji University — *Shanghai, China*
Postdoctoral Research Fellow at the College of Transportation — *2021 – 2024*

- Advisor: Professor Jian John Lu & Professor Jinsong Qian

EDUCATION

Tongji University — *Shanghai, China*
Ph.D. in Transportation Engineering — *2016 - 2021*

- Graduated with Highest Distinction/Outstanding Graduate/Honors thesis
- Dissertation: Automatic Distress Detection and Classification Algorithms Using 3-D Pavement Images
- Advisor: Professor Jian John Lu

Tianjin University of Science & Technology — *Tianjin, China*
B.Eng. in Measurement and Control Technology & Instrumentation — *2012 - 2016*

- Graduated with Highest Distinction/Outstanding Graduate
- Undergrad research advisor: Professor Yizhong Wang (Dean of the College of ECE)

RESEARCH GRANTS

Modernization and Web-Based Implementation of the Illinois Pavement Feedback System https://trid.trb.org/View/2677555. — *Champaign, IL*
Sponsor: Illinois Department of Transportation, ICT-R27-288 ($653K) — *Mar 2026 - Feb 2028*

- PI: **Hong Lang**; Co-PI: Imad L. Al-Qadi

Intelligent Infrastructure Distress Detection Using Bidirectional 3-D Triangulation and Multi-Information Fusion *Shanghai, China*

Sponsor: National Natural Science Foundation of China, P.62206201 ($30K) *2022 - 2024*

- PI: **Hong Lang**
- Proposed a method for absolute elevation measurement of surfaces using bidirectional 3-D laser imaging.

Integrated Active Road Traffic Safety Prevention and Control Technology *Shanghai, China*

Sponsor: National Key Research and Development Project of China, P.R2017YFC0803902 ($660K) *2018 - 2023*

- PI: Jian John Lu; Co-PI: **Hong Lang**, Ke Wang
- Proposed an algorithm for road traffic behaviour extraction using unmanned aerial vehicle (UAV).

Key Technologies and Equipment for Vehicle-Mounted Lightweight and Rapid Inspection of Highway Operational Safety Incidents *Shanghai, China*

Sponsor: Key Research and Development Project of Henan, P.261111242600 ($47K) *2024 - 2026*

- PI: Jinsong Qian; Co-PI: **Hong Lang**

Pavement Distress Mechanism of Spatial Domain Generalization by Multi-Data Fusion *Shanghai, China*

Sponsor: China Postdoctoral Science Foundation, P.2023M732644 ($12K) *2023 - 2024*

- PI: **Hong Lang**
- Analyzed the imaging mechanisms of pavement distresses based on multi-view data.

Development and Application of an AI-Based Pavement Inspection Data System *Shanghai, China*

Sponsor: Shanghai Science and Technology Commission, P.23692118200 ($7,500) *2023 - 2024*

- PI: **Hong Lang**
- Compiled a report on global research trends in digital road technology.

Bridge Differential Settlement Detection Based on Inertial Navigation *Shanghai, China*

Sponsor: Jiangsu Key Laboratory of Transportation Safety, P.TTS2021-03 ($4,500) *2021 - 2023*

- PI: **Hong Lang**
- Verified the detection results through manual surveys of 89 bridges.

Intelligent Monitoring and Analysis of Asphalt Pavement Service Condition *Shanghai, China*

Sponsor: Henan Provincial Department of Transportation, P.2023-1-1 ($60K) *2023 - 2025*

- PI: **Hong Lang**
- Achieved rapid detection of subsidence and asset-related defects using unorganized MLS point clouds.

High-Resolution 3D Runway Damage Detection for Airport Pavement Monitoring *Shanghai, China*

Sponsor: Tongji University, P.2023-03 ($250K) *2023 - 2024*

- PI: **Hong Lang**
- Developed a high-precision 3-D detection system with 1-mm resolution across 8,192 pixels.

SUBMITTED PROPOSALS (★ Awarded)

Illinois BMD Data Foundation and AI-Assisted AC Mixture Evaluation *IDOT-IDEA STATEMENT*

Illinois Department of Transportation *Submitted Sep 2026*

Higher-Speed Vision-Based Pavement Friction Estimation for Network-Level Safety Screening - Phase II *IDOT-IDEA STATEMENT*

Illinois Department of Transportation *Submitted Sep 2026*

Impact of Macrotexture and Water Film Thickness on Wet-Pavement Safety *FHWA*
Federal Highway Administration *Submitted July 2026*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Yanfeng Ouyang; Partner: John Senger (IDOT)

Evaluation of Non-Contact Friction and Microtexture Measurement Systems for Pavement Safety Analysis *FHWA*
Federal Highway Administration *Submitted July 2026*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Yanfeng Ouyang; Partner: John Senger (IDOT)

Modernization and Web-Based Implementation of the Illinois Pavement Feedback System ★ *IDOT*
Illinois Department of Transportation *Submitted Jan 2026*

- PI: Dr. **Hong Lang**;

Artificial Intelligence (AI) Analyses of Balanced Mix Design (BMD) Mixtures *WisDOT*
Wisconsin Department of Transportation- Wisconsin Highway Research Program *Submitted Jan 2026*

- PI: Dr. Jaime Hernandez; Co-PI: Dr. Imad L. Al-Qadi; Co-PI: **Hong Lang**

Surface Safety Readiness for CAVs on Rural Roads under Adverse Weather ★ *USDOT(CCAT)*
Center for Connected And Automated Transportation: Safety, Mobility *Submitted Nov 2025*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Yanfeng Ouyang; Co-PI: Dr. Alireza Talebpour

Unified Distress Severity Framework and Enhanced CRS Calculation Models for Illinois Pavements ★ *IDOT*
Illinois Department of Transportation: Condition Rating Survey Distress Parameters *Submitted Oct 2025*

- PI: Dr. Yanfeng Ouyang; Co-PI: Dr. Imad L. Al-Qadi;

Ensuring Resiliency of Asphalt Pavement in Real-Time during Construction Using Ground-Penetrating Radar (GPR) *USDOT*
U.S. Department of Transportation: ARPA-I Ideas and Innovation Challenge *Submitted Sept 2025*

- PI: Dr. Imad L. Al-Qadi;

Freeze–Thaw Informed Decision-Support Tool for Minnesota's Spring Load Restrictions *MnDOT*
Minnesota Department of Transportation: Improved seasonal load limits *Submitted Aug 2025*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Ashish Sharma (climate scientist)

Demonstration of an Innovative Technique to Eliminate Road Surface-Related Fatalities *USDOT*
U.S. Department of Transportation: Safe Streets and Roads for All (SS4A) *Submitted Jun 2025*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Yanfeng Ouyang and Dr. Mani Golparvar-Fard

PROJECTS

Roller-Mounted GPR for Asphalt Pavement Quality Control and Assurance *Champaign, IL*
Sponsor: Federal Highway Administration, Project 26-3554-57413 ($100K) *Jul 2024 - July 2025*

- PI: Dr. Imad L. Al-Qadi
- Co-developed a real-time **ground-penetrating radar (GPR) tool** for vehicle-mounted and road roller applications to evaluate pavement density, moisture content, and compaction degree.

Automated and Contactless Identification of Asphalt Pavement Surface Friction Based on Computer Vision Image-based 3D Reconstruction and Machine Learning Techniques *Champaign, IL*

Sponsor: Illinois Department of Transportation, Project R27-247 ($750K) *Jul 2024 - Jun 2026*

- PI: Dr. Imad L. Al-Qadi; Co-PI: Dr. Mani Golparvar-Fard
- Co-developed a pavement friction detection system based on Structure-from-Motion vision techniques.
- Developed a customized SAINT tabular DL model for friction prediction, achieving 6% higher accuracy than XGBoost and RF.

Using Advanced Binder Rheological Parameters to Predict Cracking Potential of Hot-Mix Asphalt Mixtures with Modified Binders *Champaign, IL*

Sponsor: Illinois Department of Transportation, Project R27-250 ($540K) *Oct 2024 - Sep 2025*

- PI: Dr. Imad L. Al-Qadi
- Developed an AI-driven mix design tool using 15 years of IDOT cracking and rutting data, automating binder and aggregate optimization to reduce time, cost, and material use. ICT Optimal Mix Design Tool.
- Provided guidelines for blending and thresholds for softener-modified binders, along with a model for informed material selection, reducing premature failure risk, and enabling cost-effective strategies. Asphalt Binder Selection Guidance Tool.

Update of Traffic Factor Equations for IDOT Mechanistic-Empirical Pavement Design *Champaign, IL*

Sponsor: Illinois Department of Transportation, Project R27-277 ($489K) *Dec 2024 - Present*

- PI: Dr. Imad L. Al-Qadi
- Performed clustering analysis on Weigh-in-Motion axle-load spectra and developed a physics-informed pavement–tire interaction model.

Optimizing the Use of Local Aggregates in SMA Accelerated Pavement Testing *Champaign, IL*

Sponsor: Illinois Department of Transportation, Project R27-216 *Oct 2024 - Dec 2025*

- PI: Dr. Imad L. Al-Qadi
- Developed a line-laser rut extraction algorithm to analyze pavement structural response under continuous loading.
- Conducted comprehensive engineering analysis across six SMA test sections under different loading configurations.

PATENTS, REPORTS AND BOOKS

Patents

Road 3D Reconstruction Using Primary & Auxiliary LiDAR with Integrated Navigation *China*

China. Patent Pending. Author: **H.Lang.**, Z.Zou, X.T.Chen, J.S.Qian. *2024*

Road Roughness Detection with 3D Structured Light and Vibration Fusion *China*

China. Patent Authorized. ZL202410497318.9.Author: **H.Lang.**, Z.Chen, J.S.Qian. *2024*

Pavement 3D Detection Using a Bidirectional Triangulation System *China*

China. Patent Authorized. ZL202210445208.9. Author: **H.Lang.**, Y.Yuan., J.Chen. *2023*

A Pavement Rutting Anomaly Detection System and method based on 3D Triangulation *China*

China. Patent Pending. Author: J.M.Ling., **H.Lang.**, J.S.Qian. *2023*

A Multidimensional Pavement Distress Data Processing Method *China*

China. Patent Authorized. Author: **H.Lang.**, S.Ding., Y.Y.Xing. *2022*

A 3-D Image Pavement Detection System. *China*

China. Patent Authorized. CN 110415232 A. Author: **H.Lang.**, J.J.Lu. *2020*

Communication Channel Extension System for an Internet of Things Gateway — *China*, 2019
China. Patent Authorized. ZL201610937604.8. Author: Y.Z.Wang, R.Zhang., **H.Lang.**.

Asphalt Pavement Distress (18,256 samples, detection/segmentation). Pavement Type Classification (53,265 samples, classification). Concrete Distress (7,414 samples, detection/segmentation). MLS point clouds(800km, 9,056 samples). — *China*, 2019
Data Sharing & Open Access
Owner: **H.Lang.**Updated on 2025.06.26. https://github.com/IP2RG.

Reports

Computer Vision-Based Contactless Estimation of Asphalt Pavement Surface Friction — *Illinois*, 2026
FHWA-ICT-26-017(R27-247). 10.36501/0197-9191/26-019.

- I.L.Al-Qadi, M.Fakhreddine, M.Golparvar-Fard, **H.Lang**, W.D.Vilamil

Optimizing the Use of Local Aggregates in Stone Mastic Asphalt — *Illinois*, 2026
FHWA-ICT-26-004(R27-216). 10.36501/0197-9191/26-004.

- I.L.Al-Qadi, J.J.García Mainier, **H.Lang**, Lara Diab, Greg Renshaw.

Using Advanced Binder Rheological Parameters to Predict Cracking Potential of Hot-Mix Asphalt Mixtures with Modified Binders — *Illinois*, 2025
FHWA-ICT-25-011(R27-250). 10.36501/0197-9191/25-011.

- I.L.Al-Qadi, A.O.Sulaiman, M.A.Uthman, **H.Lang**, J.J.García Mainier.

Standards and Books

Automatic Bridgehead Settlement Detection System — *China*, 2024
Group Standard of the China Highway and Transportation Society (CHTS). Standard Pending.

- J.J.Lu, Y.M.Jiang, S.D.Chen, **H.Lang**, et al.

Vehicle-Mounted 3-D Detection System for Pavement Distress — *China*, 2023
Group Standard of the CHTS. Standard Authorized.

- J.J.Lu, Y.M.Jiang, R.J.Cao, M.Cao, **H.Lang**, et al.

Advanced Pavement Quality Evaluation and Detection Technology — *China*, 2022
Tongji University Press Book. Book Published.

- J.J.Lu, Y.M.Jiang, **H.Lang**, S.D.Chen, Y.X.Lou.

SELECTED PUBLICATIONS(* corresponding author; † co-first author)

Asphalt Materials Engineering

[J2] **Hot-Mix Asphalt Cracking Potential Prediction from Binder Rheological Parameters.**
A.O.Sulaiman*; **H.Lang**; I.L.Al-Qadi.
2026. *Journal of Transportation Research Record*. 10.1177/03611981261479.

[J1] **Designing Asphalt Concrete Mixtures Based on BMD Requirements Utilizing A Two-Step Approach.**
H.Lang*; I.L.Al-Qadi; M.A.Uthman.
2025. *Journal of Transportation Research Record, 03611981251344912*. 10.1177/03611981251344912.

[C2] **Probabilistic Compliance Assessment for Balanced Mix Design.**
H.Lang*; I.L.Al-Qadi;

2027 Accepted. *TRB 106th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C1] **Hot-Mix Asphalt Cracking Potential Prediction from Binder Rheological Parameters.**
A.O.Sulaiman*; **H.Lang**; I.L.Al-Qadi;
2026 (Lectern Session). *TRB 105th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

Asphalt Pavement Mechanics and Design

[J4] **Data-Driven Sensor Analytics for Quantifying Viscoelastic Recovery of AC Pavements under Full-Scale Loading.**
J.G.Mainieri*; **H.Lang**; I.L.Al-Qadi.
2026 Under Review. *ASCE Journal of Computing in Civil Engineering*.

[J3] **3D Tire–Pavement Contact Stresses: Physics⊠Informed Prediction Approach.**
H.Lang*; W.D.Villamil; I.L.Al-Qadi.
2026. *International Journal of Pavement Engineering*, 27(1). 10.1080/10298436.2026.2621970.

[J2] **Smartphone-Based Image Analysis Method for Estimating Moisture Content in Subgrade Filling.**
H.Y.Ouyang; J.S.Qian*.**H.Lang**; J.K.Zhang; Y.Zhang;
2025. *International Journal of Transportation Science and Technology*. 10.1016/j.ijtst.2025.06.010.

[J1] **The Unification of Damage Evolution in Asphalt Mixtures under Various Strain Waveforms Using the Dissipated Energy Indicator.**
H.L.Chen; C.Y.Xue.**H.Lang**; Y.H.Wang; K.L.Zhuang; Z.C.Han; L.J.Sun; X.Y.Chen;
2025. *Construction and Building Materials, 493, 143214*. . 10.1016/j.conbuildmat.2025.143214.

[C3] **Prediction of Seasonal Axle-Load Spectra for Non-Weigh-in-Motion Interstates Using Parallel Factor Analysis.**
A.Singh*; **H.Lang**; I.L.Al-Qadi;
2027 Accepted. *TRB 106th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C2] **3D Tire–Pavement Contact Stresses: Physics-Informed Prediction Approach.**
H.Lang*; W.D.Villamil; I.L.Al-Qadi;
2026. *TRB 105th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C1] **Seasonal Load Spectra on Illinois Interstate Highways.**
A.Singh*; **H.Lang**; I.L.Al-Qadi;
2026. *The 11th International Conference on the Bearing Capacity of Roads, Railways and Airfields*. Ljubljana, Slovenia.

AI- LiDAR Fusion and Transportation Infrastructure Digitization

[J6] **Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds.**
X.T.Chen; **H.Lang***; Z.Zou; Y.Chen; Y.M.Jiang.
2026. *Automation in Construction*. 188: 106996. 10.1016/j.autcon.2026.106996.

[J5] **Strip Clustering-guided Semantic Segmentation for Pavement Deformation Detection from MLS Point Clouds.**
Z.Zou; **H.Lang***; Y.Chen.
2026 Under Review. *Engineering Applications of Artificial Intelligence*.

[J4] **Robust Estimation for Pavement Plane Coefficients from Road Point Cloud.**
H.Lang; Z.Zou*; H.L.Cheng; A.D.Wang; J.S.Qian.
2026 Under review. *IEEE Transactions on Geoscience and Remote Sensing*.

[J3] **LiDAR-Driven Innovations in Pavement Distress Detection: A Review.**
Y.H. Si†; **H. Lang**†; J.S. Qian*; Z. Zou.
2025. *ASCE Journal of Computing in Civil Engineering*. 40(1): 03125003. 10.1061/JCCEE5.CPENG-6662.

[J2] **Coarse-to-Refined Road Curb Segmentation From MLS Point Clouds.**
Z.Zou; **H.Lang***; J.J.Lu.
2024. *Automation in Construction*. 166: 105586. . 10.1016/j.autcon.2024.105586.

[J1] **Plane-based Global Registration Method for Pavement 3D Reconstruction Using Hybrid Solid-state LiDAR.**
Z.Zou; **H.Lang***; Y.X.Lou; J.J.Lu.
2023. *Automation in Construction*. 152: 104907. . 10.1016/j.autcon.2023.104907.

[C2] **Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds.**
X.T.Chen; **H.Lang*** (Presenter); Z.Zou; Y.Chen; Y.M.Jiang.
2026 (Lectern Session). *TRB 105th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C1] **Road Curb Extraction from Unordered Point Clouds Based on Multi-Feature Filtering.**
Y.Peng; **H.Lang***; Z.Zou; J.J.Lu.
2024. *TRB 103rd Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-24-02096.

AI- Intelligent Detection and Mechanism Analysis of Infrastructure Condition

[J18] **Automated Pavement Pothole Detection and Multi-dimensional Indicators Extraction Using Deep Learning and 3D Reconstruction.**
A.D.Wang; **H.Lang***; J.Chen; Z.Zou; Y.J.Zou; J.S.Qian.
2026. *IEEE Transactions on Intelligent Transportation Systems*. . 10.1109/TITS.2026.3665030.

[J17] **Automated Image-level Pavement Type Recognition on Cross-regional Data Using a Multi-feature FusionNet**
H.Lang; Z.Chen*; J.S.Qian*; A.D.Wang; Y.C.Peng; H.Du.
2025. *International Journal of Pavement Engineering*. 10.1080/10298436.2025.2569612.

[J16] **Active Learning for Infrastructure Distress Segmentation via Pixel-Level Difficulty Estimation.**
H.Lang; J.Chen*; H.R.Gong; Z.Chen; Q.W.Zhou; Y.J.Zou*.
2026 Under Review. *IEEE Transactions on Intelligent Transportation Systems*.

[J15] **The Two-step Method of Pavement Pothole and Raveling Detection and Segmentation Based on Deep Learning.**
A.D.Wang; **H.Lang***; Z.Chen; Y.C.Peng; J.J.Lu.
2024. *IEEE Transactions on Intelligent Transportation Systems*. . 10.1109/TITS.2023.3340340.

[J14] **Augmented Road Crack Segmentation: Learning Complete Representation to Defend Noise in Concrete Pavements.**
H.Lang; Y.Yuan*; J.Chen; S.Ding; J.J.Lu.
2024. *IEEE Transactions on Instrumentation and Measurement*. . 10.1109/TIM.2024.3378205.

[J13] **Selecting and Optimizing Distress Feature Descriptors from Multi-source Images for Pavement Distress Classification.**
A.D.Wang; **H.Lang***; Y.C.Peng; X.T.Chen; J.S.Qian*.
2025. *International Journal of Transportation Science and Technology*. 10.1016/j.ijtst.2025.05.003.

[J12] **Automatic Pixel-Level Detection of Multiple Pavement Distresses and Surface Design Features with PDSNet II.**
H.Lang; J.S.Qian; Y.Yuan*; J.Chen; Y.Y.Xing; A.D.Wang.
2024. *ASCE Journal of Computing in Civil Engineering*. 10.1061/JCCEE5.CPENG-5894.

[J11] **Automated Pavement Distress Segmentation on Asphalt Surfaces Using a Deep Learning Network.**
T.Wen; S.Ding; **H.Lang***; J.J.Lu; Y.Yuan; Y.C.Peng; J.Chen; A.D.Wang.
2023. *International Journal of Pavement Engineering*. 24(2), 2027414. 10.1080/10298436.2022.2027414.

[J10] **Automated Crack Segmentation on 3D Asphalt Surfaces with Richer Attention and Hybrid Pyramid Structures.**
S.Ding; **H.Lang***; J.Chen; Y.Yuan; J.J.Lu.
2023. *International Journal of Pavement Engineering*. 24(1), 2246097. 10.1080/10298436.2023.2246097.

[J9] **PCDNet: Seed Operation–Based Deep Learning Model for Pavement Crack Detection on 3D Asphalt Surface.**
T.Wen; **H.Lang***; S.Ding; J.J.Lu; Y.Y.Xing.
2022. *ASCE Journal of Transportation Engineering, Part B: Pavements*. 148(2): 04022023. 10.1061/JPEODX.0000367.

[J8] **The Improvement of Automated Crack Segmentation on Concrete Pavement with Graph Network.**
J.Chen; Y.Yuan; **H.Lang***; S.Ding; J.J.Lu.
2022. *Journal of Advanced Transportation*. 10.1155/2022/2238095.

[J7] **Line-Structured Light Rut Detection of Asphalt Pavement with Markings Interference under Strong Light.**
S.Ding, Y.Y.Xing, **H.Lang**, T.Wen, J.J.Lu*.
2022. *ASCE Journal of Transportation Engineering, Part B: Pavements*, 148(2), 04022007. 10.1061/JPEODX.0000341.

[J6] **Pavement Cracking Detection and Classification Based on 3D Image Using Multiscale Clustering Model.**
H.Lang; J.J.Lu; Y.X.Lou*; S.D.Chen.
2021. *ASCE Journal of Computing in Civil Engineering*, 34(5): 04020034. 10.1061/(ASCE)CP.1943-5487.0000910.

[J5] **A Review of Pavement Distress Detection Based on Machine Learning Methods.**
Z.Zou; J.Chen; **H.Lang***; X.F.Wang; C.G.Wan; S.Ding; J.J.Lu.
2025. *Journal of Transport Information and Safety (in Chinese)*. 43(2): 154-168. 10.3963/j.jssn.1674-4861.2025.02.016.

[J4] **Rutting Abnormality Analysis Method for 3D Asphalt Pavement Surfaces Based on Semantic Segmentation Model.**

A.D.Wang; **H.Lang***; S.Ding; J.J.Lu; X.G.Hong; T.Wen.
2023. *Journal of South China University of Technology (in Chinese)*. 51(1): 134-144. 10.12141/j.issn.1000-565X.210778.

[J3] **Multi-distress Detection Method for Asphalt Pavements Based on Multi-branch Deep Learning.**
J.Chen; Y.Yuan; **H.Lang***; T.Wen; S.Ding; J.J.Lu.
2023. *Journal of Southeast University (in Chinese)*. 53(1): 123-129. 10.3969/j.issn.1001-0505.2023.01.015.

[J2] **3D Pavement Crack Detection Method Based on Deep Learning.**
H.Lang; T.Wen; J.J.Lu*; S.Ding; S.D.Chen.
2021. *Journal of Southeast University (in Chinese)*. 51(1): 53-60. 10.3969/j.issn.1001-0505.2021.01.008.

[J1] **Asphalt Pavement Rutting Anomaly Inspection Method Considering 3D Characteristics of Distress.**
H.Lang; J.J.Lu*; S.D.Chen; Y.X.Lou.
2020. *Journal of Southeast University (in Chinese)*. 50(3): 454-462.

[C8] **Lightweight Joint Classification of Pavement Material and Distress Presence Using a Dual-Branch ConvNeXt.**
Z.Chen;**H.Lang***;J.S.Qian;
2027 Accepted. *TRB 106th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C7] **Credibility-Guided Multi-Source Self-Training for Pavement Distress Detection.**
Z.Chen;**H.Lang***;Y.Y.Xing;
2027 Accepted. *TRB 106th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C6] **Intelligent Pixel-Level Segmentation of Pavement Sealed Cracks Using CycleGAN-Based Domain Adaptation.**
Y.M.Li; J.Chen; **H.Lang***; J.S.Qian.
2025. *TRB 104th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-25-02916

[C5] **Efficient Data Selection Based on Difficulty-Aware Module for Pavement Distress Segmentation.**
J.Chen; Y.Yuan; **H.Lang***; S.Ding; J.J.Lu.
2024. *TRB 103rd Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-24-03862.

[C4] **The Two-step Method of Pavement Pothole and Raveling Detection and Segmentation Based on Deep Learning.**
A.D.Wang. **H.Lang***; Z.Chen; Y.C.Peng; J.J.Lu.
2024. *TRB 103rd Annual Meeting of the Transportation Research Board*. Washington, D.C., USA . NO. TRBAM-24-02687.

[C3] **Automatic Distress Segmentation on Asphalt Pavement Based on Deep Learning with Multiple Branches.**
J.Chen; T.Wen; S.Ding; Y.Yuan; **H.Lang***; J.J.Lu.
2022. *TRB 101st Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-22-01850.

[C2] **Automated Rutting Abnormality Detection on 3D Asphalt Pavement Surfaces Based on Disease Seeds.**
H.Lang; Y.C.Peng; J.J.Lu*.
2020. *TRB 99th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO.20-05363.

[C1] **A Review of 3D Pavement Automatic Measurement System.**
H.Lang*; Y.X.Lou; J.J.Lu*; Y.Chen.
2018. *17th COTA International Conference of Transportation Professionals*.3D Pavement System.

AI- Safety and Sustainability in Transportation Systems

[J7] **Data-Driven Network-Level Pavement Friction Prediction Using SCRIM Data: A Large-Scale Comparison of Tree-Based and Transformer-Based Tabular Models.**
M.Fakhreddine; **H.Lang**; M.Golparvar-Fard*, I.L.Al-Qadi.
2026 Under Review. *ASCE Journal of Computing in Civil Engineering*.

[J6] **Optimization of Water-road Freight Transportation Routes for Reduced Fuel Consumption and Traffic Risk.**
T.R.Zhang; **H.Lang**; Y.B.Chen*; A. Moeinaddini*; Y.Zou.
2025. *IET Intelligent Transport Systems, 19(1), e70078*. 10.1049/itr2.70078.

[J5] **A Dynamic Dual Optimization Method for Safe Lane Changing in Autonomous Driving.**
J.C.Jiang; **H.Lang**; Y.B.Chen*; Y.Zhang; Y.Wang; H.Zhang; Y.Zou*.
2025. *Transportmetrica A: Transport Science, 1-33*. 10.1080/23249935.2025.2497887.

[J4] **What Is the Impact of a Dockless Bike-Sharing System on Urban Public Transit Ridership.**
H.Lang; S.Zhang; K.Fang; Y.Xing*; Q.Xue.
2023. *Sustainability, 15(14), 10753*. 10.3390/su151410753.

[J3] **A Portable Measurement System for Pavement Surface Macrotexture–a Case Study in China.**
Y.X.Lou; S.D.Chen; J.J.Lu*; **H.Lang**.
2022. *International Journal of Pavement Engineering*. 23(12), 4136-4148. 10.1080/10298436.2021.1935939.

[J2] **The Situation of Hazardous Materials Accidents during Road Transportation in China from 2013 to 2019**
S.X.Zhu; S.W.Zhang; **H.Lang***; C.Jiang; Y.Xing.
2022. *International Journal of Environmental Research and Public Health, 19(15), 9632*. 10.3390/ijerph19159632.

[J1] **Traffic Video Significance Foreground Target Extraction in Complex Scenes.**
H.Lang; S.Ding; J.J.Lu*; X.L.Ma.
2019. *Journal of Image and Graphics (in Chinese)*. 24(01):50-63. 10.11834/jig.180313.

[C5] **Aviation Accident Report Causality Extraction Based on Transformer with BERT embeddings.**
Z.Chen; **H.Lang**; Y.Y.Xing*; L.Wang; S.W.Zhang.
2025 (Lectern Session). *TRB 104th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA.

[C4] **Automatic Pavement Type Recognition for Image-based Data Using a Multi-Feature Fusion Network.**
Z.Chen; **H.Lang***; A.D.Wang; Y.Peng; Y.Y.Xing.
2024. *TRB 103rd Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-24-02961.

[C3] **Smart Pavement: An Attention-Based Classification Model for Road Pavement Material.**
Y.Yuan; Q.W.Xue; **H.Lang***; J.Zhu; J.Chen; Y.Peng
2022. *5th KES International Symposium on Smart Transportation Systems*. 133-140. 10.1007/978-981-19-2813-0-14.

[C2] **Rutting Abnormality Detection and Correction on 3D Asphalt Pavement Surfaces Using Segmentation Model.**
A.D.Wang; S.Ding; **H.Lang***; T.Wen; J.J.Lu.
2022. *TRB 101st Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. NO. TRBAM-22-01795.

[C1] **Development of A Portable Pavement Surface Macrotexture Measurement System Applied in China.**
H.Lang, Yuexin Lou; Jian John Lu*.
2019. *TRB 98th Annual Meeting of the Transportation Research Board*. Washington, D.C., USA. No. 19-03777.

PROFESSIONAL EXPERIENCE

2024-2024	**Research Associate Professor**, Transportation Research Institute of Tongji University
2021-2024	**Senior Research Scientist**, Shanghai PRES Road Traffic Technology Co., Ltd, Supervisor: Dr. Shendi Chen
2017-2021	**Research Scientist**, Shanghai PRES Road Traffic Technology Co., Ltd, Supervisor: Dr. Jian John Lu
2017-2021	**Research Assistant**, Tongji University, Supervisor: Dr. Jian John Lu
2014-2016	**Undergraduate Research Assistant**, Tianjin University of Science & Technology, Supervisor: Dr. Fengzhi Dai

HONORS & AWARDS

2026	**Scholarship and Teaching for Engineering Postdocs (STEP)**, Grainger College of Engineering, University of Illinois Urbana-Champaign	*800 USD*
2025	**Comprehensive Award for Advancing High-Level Research Capability**, Tongji University — for the 3-D road reconstruction work published in *Automation in Construction*	*1,500 USD*
2024	**First Prize**, Shanghai Institute Traffic Engineering Science and Technology, *Driving Behavior Spectrum Theory and Intelligent Risky Driving Identification under Full-Road Perception*	*2,000 USD*
2023	**Best Paper Award**, 2nd China Highway & Transportation Society (CHTS) Smart Transportation Annual Conference	
2022	**Two Paper Awards (Technical Innovation & Technology Exploration)**, 12th Annual Conference on Highway Maintenance and Management, CHTS	
2021	**Two Paper Awards (Technical Innovation & Technology Exploration)**, 11th Annual Conference on Highway Maintenance and Management, CHTS	
2021	**Outstanding Graduate**, Tongji University	
2019	**First Prize (6th National Ranking)**, China Graduate Mathematical Modeling Competition	*1000 USD*
2018	**Second Prize**, China Graduate Mathematical Modeling Competition	

Certifications & Licenses

2026 **FAA Remote Pilot Certificate (Part 107)**, Federal Aviation Administration (FAA) (UAS field operations)

INVITED TALKS & SEMINAR

2026. *Road Alignment Extraction from MLS Point Clouds.* **Road Profile Users' Group (RPUG)**, Pittsburgh, PA, USA.

2026. *Multimodal Infrastructure Intelligence for Smarter and More Resilient Pavements.* **Tongji University**, Remote.

2026. *Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds.* **TRB 105th Annual Meeting of the Transportation Research Board**, Washington, D.C., USA.

2026. *Hot-Mix Asphalt Cracking Potential Prediction from Binder Rheological Parameters.* **TRB 105th Annual Meeting of the Transportation Research Board**, Washington, D.C., USA.

2025. *Efficient Prediction of Asphalt Concrete Mix Design Based on Performance Tests.* **ASCE International Conference on Transportation & Development (ICTD)**, Glendale, AZ, USA.

2025. *SCRIM Friction Number Prediction Using Decision Tree Algorithms.* **Annual Road Profile Users' Group (RPUG) Conference**, Dallas, TX, USA.

2025. *Aviation Accident Report Causality Extraction Based on Transformer with BERT embeddings.* **TRB 104th Annual Meeting of the Transportation Research Board**, Washington, D.C., USA.

2023. *5th Big Data and Intelligent Highway Maintenance Forum: Applications of Next-Generation AI in Highway Precision Inspection and Decision-Making.* **China Highway & Transportation Society**, Xian, China.

2022. *Smart Pavement: An Attention⊠Based Classification Model for Road Pavement Material.* Invited Talk at **5th KES International Symposium on Smart Transportation Systems**, Remote.

2021. *4th Big Data and Intelligent Highway Maintenance Forum: AI-Based 3D Pavement Inspection Technology and Applications*. **China Highway & Transportation Society**, Guiyang, China.

2021. *Development and Application of Pavement Inspection Technologies.* **Security & Privacy Seminar (World Transport Convention)**, Xian, China.

TEACHING EXPERIENCE

Fall 2026	**Pavement Design I CEE 406, Physics-informed Tire-Pavement Interaction**, Grainger College of Engineering, University of Illinois Urbana-Champaign.	*Guest Lecture*
Feb 2027	**Artificial Intelligence II: Large Language and Vision-Language Models**, Rantoul High School CS Visit, Illinois Center for Transportation.	*Upcoming Teaching/Outreach*
2025-2026	**Teaching for Engineering Postdocs (24-hour Training)**, Grainger College of Engineering, University of Illinois Urbana-Champaign	*Outstanding Participant*
Fall 2025	**Multimodal Infrastructure Intelligence: Measurement, Data, Method (2-hour Workshop)**, CEE, University of Illinois Urbana-Champaign	*Lecture*
Mar 2026	**Artificial Intelligence**, Rantoul High School CS Visit, Illinois Center for Transportation.	*Lecture*
Spring 2024	**Smart Transportation & Digital Highway Workshop: Digital Empowerment for Intelligent Highway Operations and Applications**, College of Transportation, Tongji University	*Instructor*
Fall 2023	**High-Quality Highway Construction and Maintenance Development Training: Digital Empowerment for Intelligent Highway Operations—Theory, Technology, and Practice**, College of Transportation, Tongji University	*Instructor*
Spring 2022	**Transportation Engineering (TE)-Intelligent Sensing and Detection Technologies**, College of Transportation, Tongji University	*Co-Instructor*

MENTORING

2026- **Gaurav Bhusal**, Worked on R27-288 IPFS project, UIUC M.S. Student

2025- **William Villamil Martinez**, Worked on R27-277 WIM project, UIUC Ph.D. Student

2024-2025 **AbdulGafar Sulaiman**, Worked on R27-250 Binder project, UIUC Ph.D. Student

2024- **Aditya Singh**, Worked on R27-277 WIM project, UIUC Ph.D. Student

2024-2026 **Mohammad Fakhreddine**, Worked on R27-247 Friction project, UIUC Ph.D. Student

2024-2025 **Javier García Mainieri**, Worked on R27-216 SMA-APT project, UIUC Ph.D. Student

2024-2026 **Yihan Chen**, Worked on FHWA GPR MATLAB tool, UIUC Ph.D. Student

2020-2024 **Zheng Zou**, Worked on LiDAR Fusion Project, Tongji Ph.D. Student, now an assistant professor at China Three Gorges University

2022-2024 **Yuan Peng**, Worked on LiDAR Fusion Project, Tongji M.S. Student

2023-2024 **Xinting Chen**, Worked on LiDAR Fusion Project, Tongji M.S. Student

2020-2022 **Tian Wen**, Worked on AI- Intelligent Infrastructure Detection, Tongji M.S. Student

2020-2023 **Shuo Ding**, Worked on AI- Intelligent Infrastructure Detection, Tongji Ph.D. Student

2020-2024 **Ye Yuan**, Worked on AI- Intelligent Infrastructure Detection, Tongji Ph.D. Student, now an Assistant Professor, Nanjing University of Aeronautics and Astronautics

2020-2024 **Jiang Chen**, Worked on AI- Intelligent Infrastructure Detection, Tongji Ph.D. Student

2020-2024 **Aidi Wang**, Worked on Intelligent Infrastructure Monitoring, Tongji Ph.D. Student, later a visiting Ph.D. student at the University of Cambridge

2021-2024 **Zhen Chen**, Worked on Intelligent Infrastructure Monitoring, Tongji Ph.D. Student

2023-2024 **Yiming Li**, Worked on AI- Intelligent Infrastructure Detection, Tongji Undergraduate Student, now a Ph.D. student at Tongji University

2023-2024 **Junyi Hu**, Worked on AI- Intelligent Infrastructure Monitoring, Tongji Undergraduate Student, later enrolled in a master's program at ETH Zurich

2023-2024 **Li Chen**, Worked on Intelligent Infrastructure Monitoring, Tongji Undergraduate Student

PROFESSIONAL SERVICE

Program Committee

2025 **Visiting Research Scientist Search Committee, CEE, UIUC**, Committee Member

2024 **Visiting Research Scientist Search Committee, CEE, UIUC**, Committee Member

2024 **Guest Editor: Journal of Transport Information and Safety**, Special Issue on AI-Driven Proactive Traffic Safety Using Road Holographic Perception

2024 **China Highway & Transportation Society**, Expert Committee

2023 **Shanghai Highway & Transportation Society**, Member

External Reviewer [Journal]

Transportation Research Part C, Automation in Construction, IEEE Transactions on Intelligent Transportation Systems, Engineering Applications of Artificial Intelligence, Neurocomputing, Reliability Engineering & System Safety, ASCE Journal of Computing in Civil Engineering, International Journal of Pavement Engineering, Construction and Building Materials, Structural Control and Health Monitoring, Measurement, Case Studies in Construction Materials, IEEE Open Journal of Intelligent Transportation Systems, Computers and Electrical Engineering, Journal of Advanced Transportation, Journal of Zhejiang University (Engineering Science).

External Reviewer [Conference]

Transportation Research Board (TRB), International Conference on Accelerated Pavement Testing (APT), ASCE International Conference on Transportation & Development (ICTD), COTA International Conference of Transportation Professionals (CICTP).

MEDIA PRESS

2026 **Researchers develop SMA using Illinois aggregate**, Illinois Center for Transportation
2026 **Fakhreddine receives SCG Global Impact Prize**, Illinois Center for Transportation
2025 **AI tool for optimal mix design of flexible roadways**, Illinois Center for Transportation
2025 **Cracking the code to predict modified asphalt binder performance**, Illinois Center for Transportation

COMPUTER SKILLS AND ELECTRICAL FUNDAMENTALS

Proficiency in C++ (13 years of experience), C#, VB.NET, Python, NI LabVIEW, HALCON, MATLAB, SQL, and SPSS.

Instrumentation experience includes measurement and control circuits, sensors, automatic control, error analysis, and analog and digital circuits. Hardware experience includes Automation Technology 3-D cameras, DJI LiDAR, Bynav INS, Teledyne DALSA cameras, NI and MCC data-acquisition systems, point lasers, accelerometers, IMUs, and encoders.